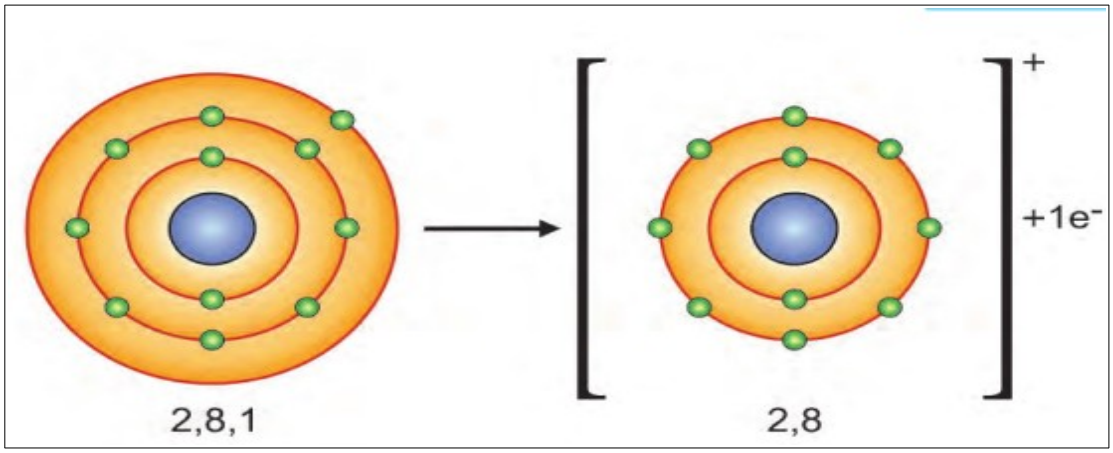
Chemical Bonding

The arrangement of eight electrons in the outermost shell, as in noble gases, is known as octet configuration.

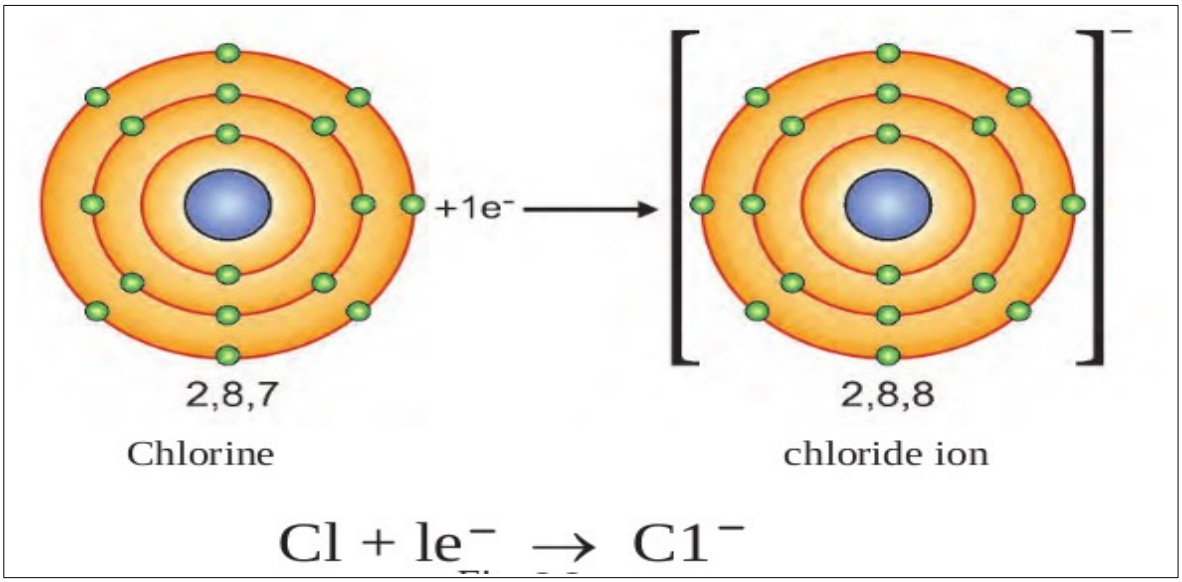
The force of attraction that holds together the constituent particles of a compound is called a chemical bond.

Molecule	Number of atoms
Oxygen (O_2)	2
Water (H_2O)	3
Nitrogen (N_2)	.....2.....
Helium (He)	....1.....
Methane (CH_4)	.....5.....
Sugar ($C_{12}H_{22}O_{11}$)	....45...

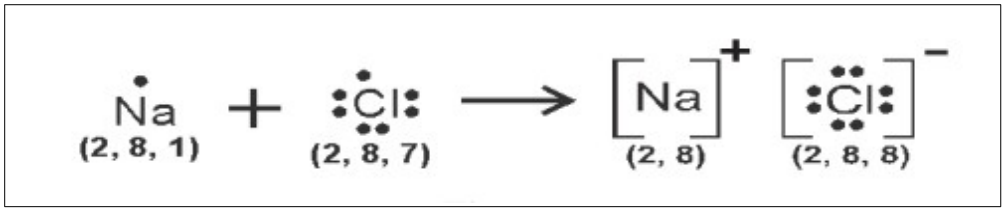
The amount of energy required to remove the most loosely bound electron from the outermost shell of an isolated gaseous atom of an element is called its ionisation energy.



$$Na \rightarrow Na^{+} + 1e^{-}$$

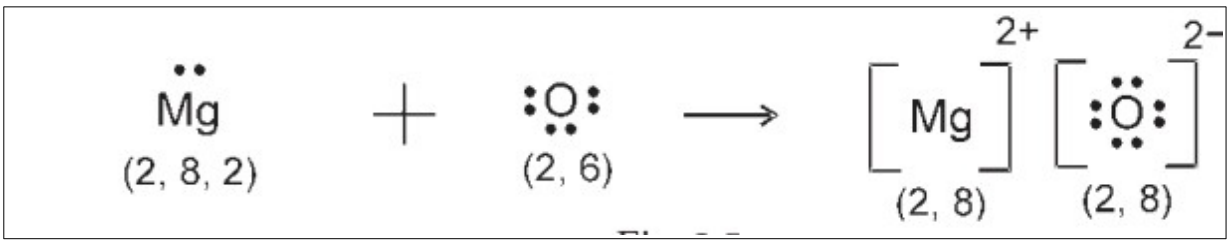


Electron dot diagram



	Sodium		Chlorine	
	Before chemical reaction	After chemical reaction	Before chemical reaction	After chemical reaction
Electron configuration	2,8,1	2,8	2,8,7	2,8,8
Number of electrons	11	10	17	18
Number of protons	11	11	17	17
Charge	o	+1	o	-1

The electrostatic force of attraction that holds together the oppositely charged ions in an ionic compound is called ionic bond. Ionic bond is also known as electrovalent bond.

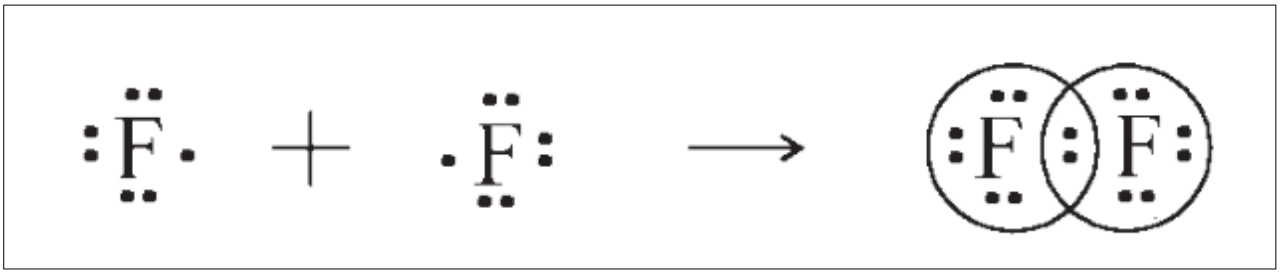


The compounds that are formed by ionic bonding are known as ionic compounds or electrovalent compounds.

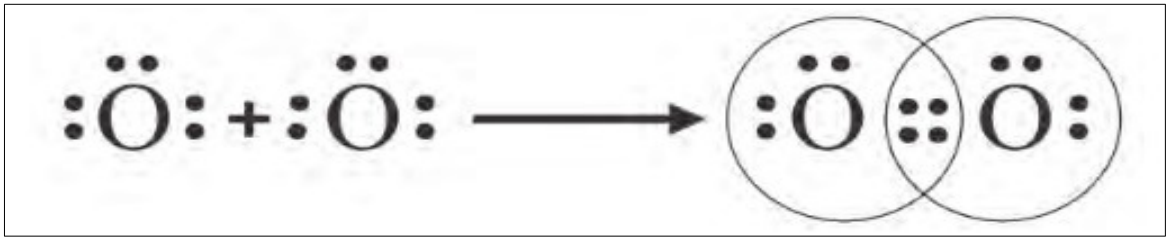
<u>Characteristics of ionic compounds</u>

- **Ionic compounds are generally soluble in polar solvents such as water.**
- **They are non volatile and hard.**
- **They exist as crystals in solid state.**
- **Generally, they have high melting points and boiling points.**

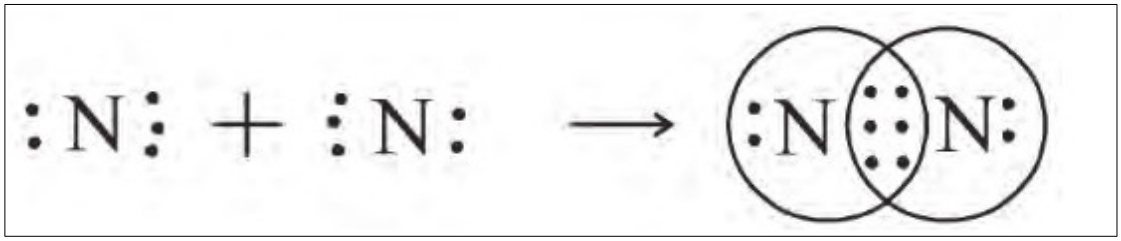
The chemical bond formed as a result of the sharing of electrons between the combining atoms is called a covalent bond. The covalent bond formed by the sharing of one pair of electrons is a single bond.



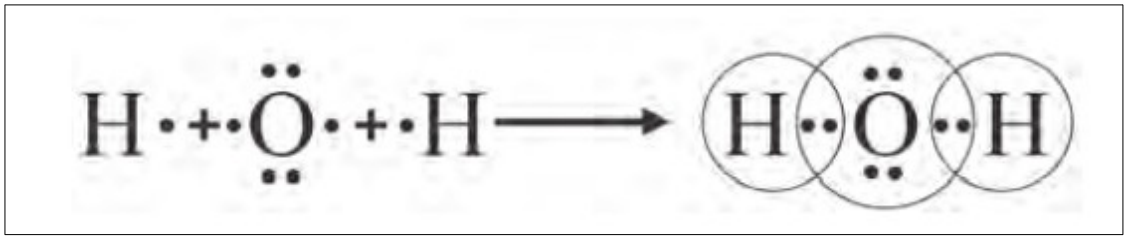
A double bond is the covalent bond formed by the sharing of two pairs of electrons.



A triple bond is the covalent bond formed by the sharing of three pairs of electrons.



Bonding in water



Compounds formed by covalent bonding are called covalent compounds. When non-metals combine together, usually covalent compounds are formed.

<u>General characteristics of covalent compounds</u>

• Covalent compounds exist in solid, liquid and gaseous states.
• They are generally insoluble in water.
• They are soluble in organic solvents like kerosene, carbon tetrachloride, benzene etc.

The relative ability of an atom to attract the shared pair of electrons between the covalently bonded atoms towards itself is called electronegativity.
If the difference in electronegativity values between the constituent elements in a compound is 1.7 or more, it generally shows ionic character and if it is less than 1.7, it shows covalent character.

The number of electrons lost, gained or shared by an atom during a chemical reaction is its valency.

Element- 1		Element - 2		Chemical formula of the compound
Name	Valency	Name	Valency	
Potassium (K)	1	Oxygen	2	.K2O.
Zinc (Zn)	2	Chlorine	1	...ZnCl2
Carbon (C)	4	Chlorine	1	..CCl4
Magnesium (Mg)	2	Oxygen	2	..MgO

Compounds	Difference in electronegativity of constituent elements	Nature of compound
Sodium chloride (NaCl)	3.16 - 0.93 = 2.23	Ionic
Hydrogen chloride (HCl)	3.16 - 2.20 = 0.96	Covalent
Sodium oxide (Na_2O)	3.44-3.16=0.28	covalent
Calcium chloride ($CaCl_2$)	3.16-1=2.16	ionic
Methane (CH_4)	2.55-2.20=0.35	covalent
Magnesium fluoride (MgF_2)	3.98-1.31=2.67	ionic

Compound	Constituent elements	Atomic number	Electron configuration	The number of electrons shared or transferred by each atom	Valency
NaCl	Na	11	2, 8, 1	1	1
	Cl	17	2, 8, 7	1	1
MgO	Mg	12	2,8,2	2	2
	O	16	2,8,6	2	2
HF	H	1	1	1	1
	F	9	2,7	2	2
CCl_4	C	6	2,4	4	4
	Cl	17	2,8,7	2	2
BeF_2	Be	4	2,2	2	2
	F	9	2,7	2	2
H_2O	H	1	1	1	1
	O	8	2,6	2	2

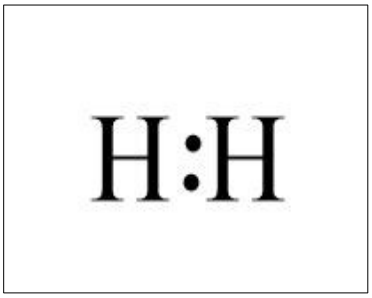
The sum of the charges of the positive ions and the negative ions in a salt will be zero.

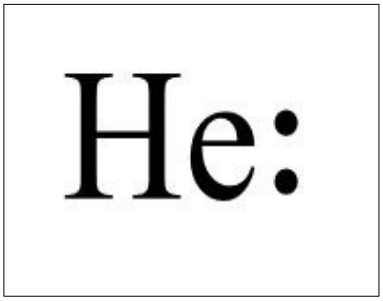
Negative ion in acid	Basicity	Chemical formula of acid
Cl^-	1	HCl
SO_4^{2-}	2	H_2SO_4
PO_4^{3-}	3	H3PO4
NO_3^-	1	HNO3
CO_3^{2-}	2	H2CO3
SO_3^{2-}	2	H2SO3

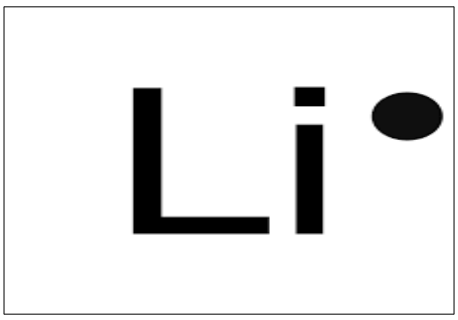
Positive ion	Negative ion	Name of salt	Chemical formula
Mg^{2+} (Magnesium ion)	Cl^- (Chloride ion)	magnesium chloride	MgCl2
Mg^{2+} (Magnesium ion)	SO_4^{2-} (Sulphate ion)	magnesiumsulphate	MgSO4
Ca^{2+} (Calcium ion)	CO_3^{2-} (Carbonate ion)	calciumcarbonate	CaCO3
NH_4^+ (Ammonium ion)	Cl^- (Chloride ion)	amoniumchloride	NH4Cl
NH_4^+ (Ammonium ion)	PO_4^{3-} (Phosphate ion)	amoniumphosphate	NH4PO4
Ca^{2+} (Calcium ion)	PO_4^{3-} (Phosphate ion)	calciumphosphate	Ca3(PO4)2
Na^+ (Sodium ion)	SO_4^{2-} (Sulphate ion)	sodiumsulphate	Na2SO4

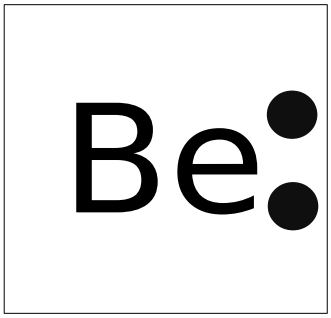
amoniumphosphate—$(NH_4)_3PO_4$

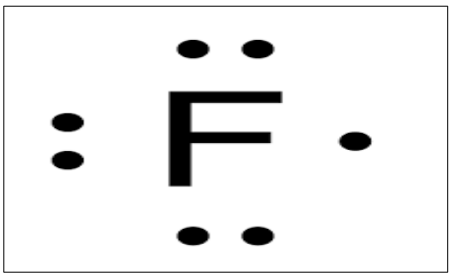
Draw the electron dot diagram of hydrogen (H), helium (He), lithium (Li), beryllium(Be) and fluorine (F).



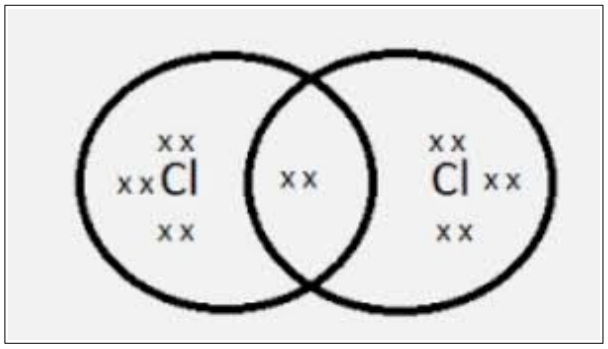








Represent the covalent bond in chlorine molecule using symbols.



Represent the formation of ionic bond in the following ionic compounds using electron dot diagram and orbit model.

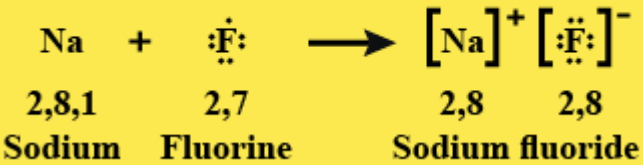
a) Sodium fluoride (NaF)

b) Sodium oxide (Na 2 O)

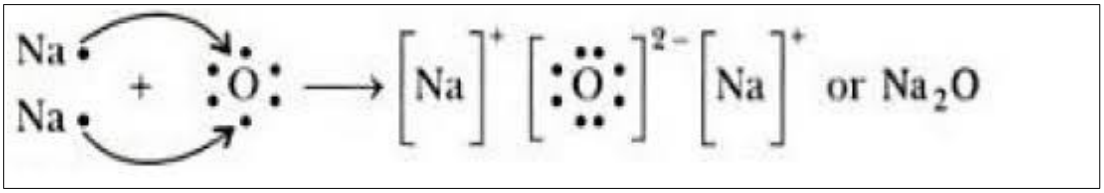
c) Magnesium fluoride (MgF 2)

d) Calcium oxide (CaO)

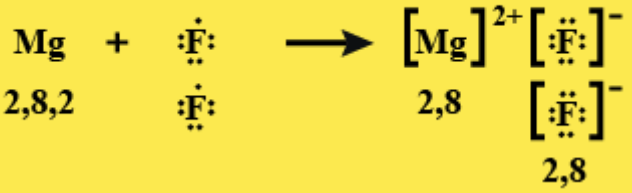
1. Sodium fluoride [NaF]

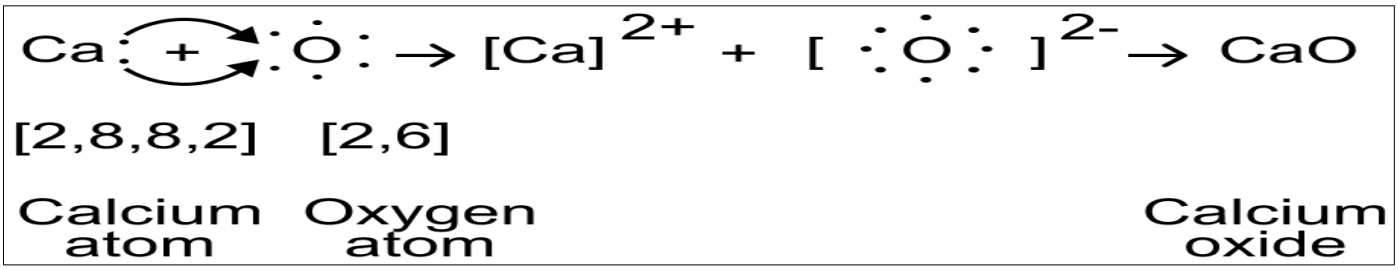


Sodium Fluorine Sodium fluoride



2. Magnesium fluoride [MgF_2]





5. Assume that calcium (Ca) and fluorine (F) combine together.

a) Complete the following table accordingly.

Element	Atomic number	Electron configuration	Number of electrons received or donated
Ca	20	2,8,10	2
F	9	2,7	1

b) Write the chemical formula of calcium fluoride.

c) Similarly, write the chemical formula of magnesium chloride and aluminium

b)CaF_2

c)$MgCl_2$ & $AlCl_3$

Some cations and anions are given in the table. Fill in the blanks.

Cation	Anion	Compound
Mg+2	Cl^-	$MgCl_2$
Na^+	F-	NaF
NH_4^+	SO_4^{2-}	(NH4)2SO4
K^+	(CO3)2-	K_2CO_3

Complete the following chemical equations and answer the questions given below.
(Hint: Atomic number Mg-12, Cl-17)

$Mg \rightarrow Mg^{2+}$ + ...2e...

$Cl + 1e^{-} \rightarrow$ .....Cl-..

...Mg.. + ...Cl2... $\rightarrow MgCl_2$

(a) Identify the cation and anion in these compounds.

(b) What is the nature of the chemical bond in $MgCl_2$?

a)anion—cl cation----Mg
b)ionic bond

Complete the following table. (Hint: Atomic number F – 9, Cl – 17, O – 8, N – 7)

Molecule	Number of shared electrons	Chemical bond
F_2	1	Single bond
Cl_2	1	single bond
O_2	2	double bond
N_2	3	tripple bond

Complete the following table. (Symbols are not real)

Element	Atomic number	Electron configuration
P	12	...2,8,2.....
Q	...9...	2,7
R	10	....2,8.....
S	17	2;8;7.....

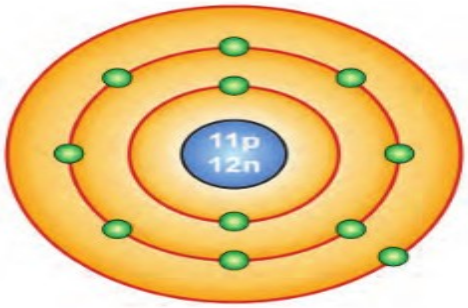
(a) Which among these is the most stable element?

(b) Which element donates electrons during chemical reactions?

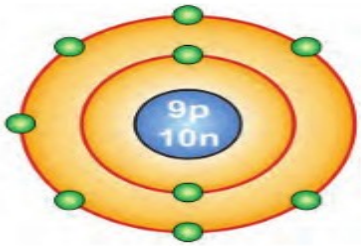
(c) Write the chemical formula of the compound formed when the elements P and S combine.

a)R b)P c)PS_2

Atom models of two elements are represented below.



Sodium



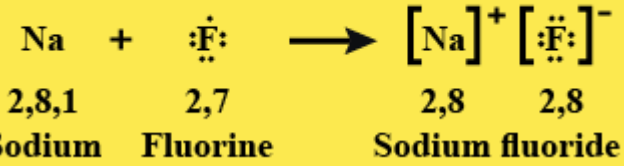
Fluorine

(a) Draw the electron dot diagram of the formation of sodium fluoride.

(b) What is the nature of chemical bond in sodium fluoride?

a)

1. Sodium fluoride [NaF]



b)ionic bond

The electron configuration of the elements P, Q, R are given below. (Symbols are not real)

P – 2,8,6

Q – 2,8,1

R – 2,8,8

(a)Which is the most stable element among these? What is the reason?

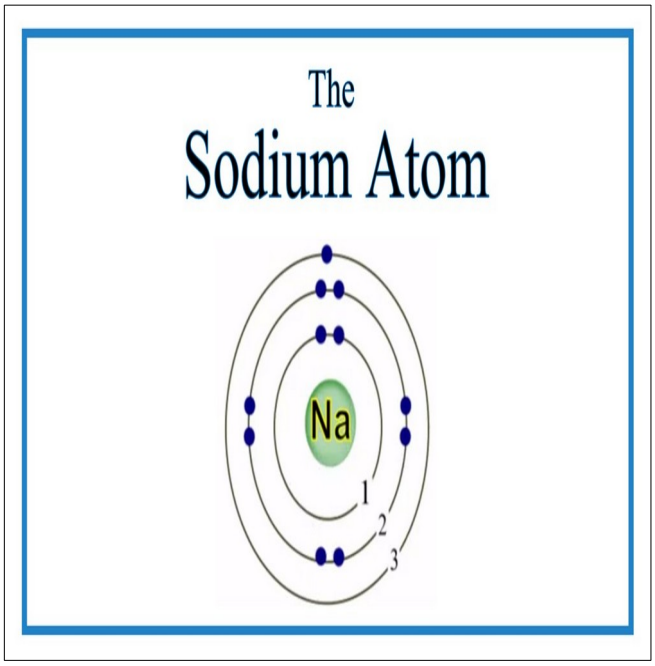
b)What is the atomic number of Q?

c)Draw the atom model of Q.

d)What are the valencies of the elements P and Q?

e)Write the chemical formula of the compound formed when P and Q combine.

a)R,because the outermost shell is filled with 8 electrons(octet)
b)11
c)



d)P=2,Q=1
e)Q_2P

. A, B, C and D are four elements (Symbols are not real). Information about them are given in the following table.

Element	Atomic number	Electronegativity
A	6	2.55
B	8	3.44
C	12	1.31
D	17	3.16

Based on these, find the type of bond in the compounds formed by the combination of the following pairs of elements.

1. C, B 2. C, D 3. A, B

1)ionic 2)ionic 3)covalent